Research Strategy and Power Estimation

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Research Aims

- **Aim 1: Inflammation is Related to Declines in Cognition and Cortical Thickness**
  - Hypothesis 1a: **Baseline** inflammation measures predict declines in memory consolidation and AD-signature cortical thickness
  - Hypothesis 1b: **Increases** in inflammation measures predict declines in memory consolidation and AD-signature cortical thickness

- **Aim 2: Inflammation and Amyloid Deposition Interact**
  - Hypothesis 2: Declines in memory and AD-signature cortical thickness are greatest when amyloid deposition and peripheral inflammation are both elevated

Data Overview

- Sample Size
  - Target 175
  - Accounting for 10% attrition
- Outcomes (Baseline – Year 1)
  - AD-Signature Cortial Thickness (continuous)
  - Cognitive measures (continuous)
- Inflammation Measures
  - Cytokines: IL-6, TNF-alpha
  - Chemokines: MCP-1, eotaxin-1
- Covariates
  - Demographics
    - Age, Sex
  - Cardiovascular risk markers
    - BMI, blood pressure, history of hypercholesterolemia
  - NSAID use
  - Immune-related health conditions
    - Crohn's disease, irritable bowel syndrome (IBS), and arthritis

Table 3 Cognitive Outcome Variables

Focus	Measures	Variables
Recall/Recognition	CVLT	20' Delay, Total Correct
		Recognition, D-Prime
	Benson Figure	15' Delay, Total Correct
Pattern Separation	MST	Proportion Correct
Long Term Consolidation	Story Recall	1-Week Delay, Total Correct

Aim 1 Analysis

- Primary models
  - Linear Regressions of Change in Outcomes (baseline – Year 1)
- Hypothesis 1a
  - Outcome Change by **Baseline Inflammation Measures**
  - Adjusting for baseline outcome
- Hypothesis 2a
  - Outcome Change by **Change in Inflammation Measures** (baseline – Year 1)
  - Adjusting for baseline outcome
- Covariates
  - Age, Sex, Variables Related to Inflammation (determined with exploratory analysis)
- Testing
  - Omnibus Test of Inflammation Measures
  - Follow-up using standardized estimates and CIs if significant
- Multicollinearity Plan
  - Fit outcome change by each inflammation measure **separately**
- Type I Error Control
  - Gatekeep with omnibus tests
  - Bonferroni correction for testing two main outcomes
    - Cortical Thickness
    - Cognitive Measures
  - $\alpha = 0.05/2 = 0.025$

Aim 1 Power

- Sample Size: $\mathrm{n} = 175$
- Type I Error Rate: $\alpha = 0.025$
- Effect Estimates
  - Omnibus test: Proportion of Variance Explained (R^2)
  - Single predictors: Correlation (ρ)
- Effects Estimated from Preliminary Data
  - R^2: 0.11 to 0.48
  - ρ: 0.26 to 0.69
- Results
  - 95% power for $R^2 = 0.11$
  - 89% power for $\rho = 0.26$
- Software
  - Omnibus test: mlrF.overall (powertools)
  - Single Predictors: corr.1samp (powertools)

R^2	Power
0.05	0.552
0.10	0.919
0.20	1.000
0.30	1.000
0.40	1.000
0.50	1.000

ρ	Power
0.15	0.400
0.20	0.665
0.25	0.868
0.30	0.966
0.35	0.995
0.40	1.000
0.45	1.000
0.50	1.000

Aim 2 Analysis

- Primary models
  - Extend Hypothesis 1b models to include **Amyloid Deposition Interaction**
- Covariates
  - Age, Sex, Variables Related to Inflammation (determined with exploratory analysis)
- Testing
  - Tests for significant interaction effects
  - Interpret inflammation-outcome relationship across levels of amyloid deposition
- Multicollinearity Plan
  - Fit outcome change by each inflammation-amyloid deposition interaction **separately**
- Type I Error Control
  - Bonferroni correction for testing two main outcomes
  - $\alpha = 0.05/2 = 0.025$

Aim 2 Power

- Sample Size: $n = 175$
- Type I Error Rate: $\alpha = 0.025$
- Dichotomous Amyloid Deposition (positive/negative)
- Assume weak correlation in amyloid-negative group ($\rho = 0.1$)
- Effect Estimates
  - $\geq 80\%$ power for large effects ($\Delta\rho \geq 0.5$) and balanced groups (30-70%)
- Software
  - corr.2samp (powertools)

	Percent with Amyloid Deposition				
$\Delta\rho$	10%	30%	50%	70%	90%
0.1	0.032	0.052	0.058	0.051	0.031
0.2	0.072	0.160	0.191	0.159	0.070
0.3	0.149	0.377	0.449	0.375	0.146
0.4	0.282	0.668	0.756	0.666	0.279
0.5	0.480	0.901	0.949	0.901	0.479
0.6	0.720	0.990	0.997	0.990	0.723
0.7	0.923	1.000	1.000	1.000	0.926
0.8	0.997	1.000	1.000	1.000	0.998